06. CONSTRUCTION OF THE PRIMITIVE VESSELS

DURATION : 13:32

STUDY SHEET

COURSE SUMMARY

This video explores the fascinating construction of primitive vessels within the embryo, highlighting the movement of fluids and the embryonic forces acting from the outside to the centre. Key concepts include the vital role of the amniotic cavity, the importance of an intercellular substance, and the four components necessary for vascular formation. The teaching also addresses the development of primitive arterial and venous systems, polarity, and the phases of assimilation and dissimilation, while emphasising how these interconnected processes form the basis of embryonic vascular networks. This knowledge is essential for any practitioner wishing to understand the anatomical and physiological foundations of human development.

CONSTRUCTION OF PRIMITIVE VESSELS

1. FLUID MOVEMENT AND EMBRYONIC FORCES

At the moment the embryo penetrates the **uterine mucosa**, all fluidic force comes from the outside and centres within the embryo. The **embryonic force** develops from the periphery towards the centre, whether fluidic, venous, or arterial. Everything converges towards the centre.

2. ROLE OF THE AMNIOTIC CAVITY AND YOLK SAC

The integration of fluids is enabled by the **amniotic cavity**. The anlage of the vascular system appears in the **splanchnopleuric mesoblast**, at the level of the wall of the **yolk sac**, the **chorion**, and the **intra-embryonic mesenchyme**. This manifests as small vascular islets in the yolk sac. The amniotic cavity allows the integration of this vitelline cavity towards the centre, marking the vitelline, placental, and neonatal stages.

3. THE FOUR ESSENTIAL COMPONENTS FOR VASCULAR FORMATION

For **kinetics** and vessel formation to occur, four components are always necessary:

1. The presence of an **intercellular substance**.

2. A **local chemical concentration**.

3. The presence of **vacuolisation**.

4. The presence of **trajectories**.

Blechmit compares this to a **rainbow**, which requires water, rain, sun, and a certain angulation. If a single element is removed, the rainbow immediately disappears. The same applies to vessels. If a trajectory or a concentration is removed, the process stops. The vascular system is in constant adaptation throughout life.

4. DEVELOPMENT OF THE VASCULAR NETWORK

A large network develops from the yolk sac, but also from the chorion. The embryo initially develops in a **"racket"** shape, with an **embryonic pedicle**. The **notochord**, located below the primitive streak, influences the neural plate. The cells of the neural plate do not grow at the same rate as the surrounding cells, leading to the development of a **groove**.

5. RESTRUCTURING OF FLOW AND VESSEL FORMATION

This groove space leads to a restructuring of the **trophic metabolic flow** within the mesoderm, in the form of bilateral fluidic trajectories. In mesodermal development, small **vacuoles** appear. This is a growth process where the fluidic mesodermal tissue is aspirated. These vacuoles grow, organising a trajectory with significant concentration and intercellular substances.

All these components allow for the formation of a vessel. These structures continue to grow to form what is called a **corrosion field**. When these fields meet, the tissue nourishing the membrane yields, allowing the progressive development of a vessel.

6. POLARITY AND PHASES OF ASSIMILATION AND DISASSIMILATION

A field of aspiration, given by this wave and methodical process, is essential. Two movements are at work: an **autochordal** movement and an underlying aspiration movement. The notochord reorganises the ectoderm into a groove, organising bilateral trajectories around the neural tube. The vacuoles meet, forming the first vessel, called the **assimilation vessel**. Cells feed on it.

A **polarity** is observed: food arrives from the ectoderm, creating an aspiration. Initially, congestion is low, but it becomes more significant in a later phase.

7. APPEARANCE OF PRIMITIVE ARTERIAL AND VENOUS SYSTEMS

This congestive phase recreates an elimination phase. An **arterial** type vessel appears through the assimilation phase (aspiration field). Then, through the return phase and the elimination of **catabolites**, a second level develops: a **descending network**.

There is thus a system of first assimilation of information, then a second plane of disassimilation. The first develops the **primitive arterial system**, the second the **secondary venous network**. These two returning fluidic trajectories are the anlage of the **cardinal vein** and the two **primitive aortas**. The venous system is more lateral and external than the primitive aortas.

8. PRIMITIVE EMBRYONIC NETWORKS

The first vascular network to develop is that of the **primitive aortas**, under the dependence of developmental movement. The uplift between the ectoderm and the endoderm aspirates from the primitive streak, which becomes the mesoderm.

The mesoderm, through the development of the neural plate, reorganises and creates trajectories with vacuolisations. These vacuolisations, through the aspiration field, create the necessary components to form a **double aorta** that will join towards the midline. Laterally and anteriorly, a return occurs in the form of trajectories of the cardinal veins.

Thus, the first embryonic network is a primitive aortic network of nutritional assimilation, and the return is the disassimilation phase. This first vascular network is therefore composed of:

1. The **primitive aortas**.

2. The network of **cardinal veins**.

The whole is integrated by the **embryonic pedicle** and the **vitelline system**. The vitelline system is crucial for the visceral system.

9. THE ROLE OF THE SPLEEN AND LIVER IN VOLUMETRIC AND VASCULAR BALANCE

The four components for all vessels are essential. Clinical cases show that volumetric changes can recreate vascular fields. For example, in an anaemic patient due to an iron problem, the **spleen**, a volume regulator, can increase blood volume by bringing back red blood cells.

The spleen can "reopen" and re-establish a concentration, reactivating old vascular networks that had atrophied. Any malformation of the spleen is linked to a malformation of the vascular system. The spleen is fundamental for rebalancing the volume of the system, by adapting its volumetry to metabolic needs.

10. SYNTHESIS OF PRIMITIVE VASCULAR COMPONENTS

The assimilation phase, organised by the notochord and neural tube, reorganises a groove and a trajectory. It includes the necessary components:

- The **trajectory**.
- The **vacuole**.
- The **metabolic concentration**.
- The presence of **intercellular substances**.

These elements give rise to a primitive vascular system of assimilation (primitive arterial system) and, through the return, to a disassimilation system (venous system). The first primitive aortas and the two cardinal veins (primitive cardinal veins) constitute this first vascular network.

The ectoderm reorganises the vascular plane, but the first aspiration field is linked to a metabolic field (permeation and infusion movements). Fluids must always be brought back to the centre.

The spleen is a great motor, but for a functional spleen, the **liver** is essential. The liver, through its growth and orientation, is the motor of organisation. It is a global (sub-diaphragmatic) and left-right organisational centre, structuring the gastro-hepato-pancreatico-splenic space. The splenic system depends on the liver.

FIG. — Lateral embryonic folding